Identifying and Rewriting Between Portuguese Varieties with Fine-tuned Language Models

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Portuguese has standardized varieties such as European Portuguese (EP) and Brazilian Portuguese (BP), yet web-scale Portuguese is dominated by BP, which can make generic Portuguese systems less reliable for EP. We study how pretrained encoder-decoder models can be adapted for both Portuguese variety identification and bidirectional EP/BP rewriting, comparing encoder-controlled and decoder-controlled formulations within a staged pipeline of supervised adaptation and reward-based continuation. Across the supervised comparisons, supervision composition produces the largest rewriting changes. GPT-Wikipedia, constructed around localized variety adaptation, gives the strongest Golden Collection rewriting results, whereas adding the Few-shot Region-aware Machine Translation (FRMT) training split sharply lowers Golden Collection performance in both directions and changes FRMT rewriting in a direction- and control-dependent manner. Control placement also depends on rewriting direction, with decoder control stronger in every BP-to-EP comparison and EP-to-BP results remaining mixed. Reward-based continuation has modest effects after GPT-Wikipedia specialization but recovers a substantial part of the Golden Collection loss after GPT-Wikipedia + FRMT specialization. These results show that Portuguese variety adaptation depends on supervision reflecting localized EP/BP rewriting and evaluation signals that distinguish necessary changes, appropriate copying, and broader paraphrase.

Additional Key Words and Phrases: Portuguese varieties, variety identification, text rewriting, encoder-decoder models, parameter-efficient fine-tuning

1 Introduction

Most Natural Language Processing (NLP) systems expose Portuguese as a single language option even though it is used through different national standards. When web-scale training data underrepresent one standard, variety-sensitive behavior can drift toward the variety that is more frequent in the data. BP is substantially more visible online than EP, so generic Portuguese systems can favor BP and serve EP less reliably [Sousa et al. 2025].

Portuguese is shared across communities with different linguistic standards, and the EP/BP contrast involves orthographic conventions, lexical choices, morphosyntactic preferences, clitic placement, verbal periphrases, and forms of address [Barreiro and Mota 2018]. These distinctions provide evidence for variety identification and determine which changes are appropriate during rewriting, where many valid EP/BP adaptations require only localized edits and some sentences require no modification.

Variety identification tests whether a system can detect subtle variety cues, while bidirectional EP/BP rewriting tests whether it can produce the requested variety while preserving the original meaning. Rewriting makes supervision quality especially important, as aligned EP/BP data must distinguish variety-specific adaptation from paraphrastic or editorial changes. The available resources differ in domain, alignment quality, construction procedure, and the transformations represented by their aligned pairs. Fine-tuning a model on these resources exposes it to rewriting behavior shaped by how supervision is constructed and combined.

The experiments fine-tune pretrained T5Gemma 2 encoder-decoder models for both tasks under a shared control-string setup. Encoder-controlled and decoder-controlled formulations differ in whether variety information is placed in the encoder input or the decoder target. Training follows a staged pipeline with large-scale supervised adaptation, targeted supervised specialization, and reward-based continuation using a group-relative objective inspired by Group Relative Policy Optimization (GRPO) [Shao et al. 2024]. A lower-cost supervised protocol-selection study fully fine-tunes the 270M-parameter model to define the protocol later used to adapt the 4B-parameter model with LoRA.

The paper makes three contributions.

- It evaluates a shared T5Gemma 2 setup for EP/BP variety identification and bidirectional rewriting under encoder-controlled and decoder-controlled formulations.
- It constructs GPT-Wikipedia as reviewed supervision for localized EP/BP adaptation and examines how supervision composition affects identification and rewriting.
- It studies staged adaptation from OpenSubtitles exposure through targeted specialization and reward-based continuation, combining aggregate evaluation with analysis of copying, required edits, and unnecessary reformulation.

Supervision composition yields the largest rewriting changes: GPT-Wikipedia aligns most closely with the conservative Golden Collection, while adding FRMT sharply lowers Golden Collection rewriting and changes FRMT performance differently across directions and control formulations. Control placement also depends on rewriting direction, with decoder control stronger in every reported BP-to-EP rewriting comparison, whereas EP-to-BP results are mixed and include settings where encoder control performs better. Stage A and Stage C effects depend on the surrounding supervision, with reward-based continuation producing its clearest Golden Collection recovery after GPT-Wikipedia + FRMT specialization. Copy analysis further shows that metrics can reward preservation but cannot fully distinguish required adaptation, acceptable paraphrase, and unnecessary reformulation.[1]

The remainder of the paper is organized as follows. Section 2 reviews the most relevant literature. Section 3 describes the data, task formulations, and training setup. Section 4 presents the quantitative and qualitative results. Section 5 discusses the main trade-offs and limitations, and Section 6 concludes the paper.

2 Related Work

This section aims to position this research within prior work on Portuguese variety processing and model adaptation. Section 2.1 reviews work on European and Brazilian Portuguese identification and rewriting, while Section 2.2 examines shared encoder-decoder modeling, staged adaptation, and reward-based continuation.

2.1 Portuguese Variety Identification and Rewriting

Early work on EP/BP identification showed that orthographic, lexical, and character-level patterns provide strong discriminative evidence. Using character n-grams and word n-gram language models, Zampieri and Gebre [2012] demonstrated the feasibility of distinguishing the two varieties from surface cues. Preda et al.

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[1] Code and reproducibility artifacts: GitHub. Reported model weights: Hugging Face. Curated data and provenance: Hugging Face.

[2024] examine whether this evidence remains reliable across different input lengths and domains by comparing surface-based classifiers with LoRA-adapted models. Their LoRA-adapted Albertina classifiers obtain the strongest in-domain results, whereas a logistic-regression bigram model performs best in the joint cross-dataset evaluation, showing that the strongest in-domain system is not also the strongest under their cross-domain comparison.

Cross-domain robustness also depends on what signals the training labels encourage a model to learn. Labels inferred from country, domain, or URL metadata can encode source-specific proxies rather than reliable variety cues [Zampieri et al. 2014]. PtBrVarId, introduced by Sousa et al. [2025], evaluates EP/BP identification explicitly across domains. Its experiments show that partial delexicalization can improve identification by reducing reliance on named entities and domain-specific lexical signals, with its strongest systems combining this procedure with BERTimbau [Souza et al. 2020]. Sousa et al. therefore provide the closest dedicated EP/BP identification reference, whereas the present setting learns identification in the same model used for rewriting. Their work provides a strong dedicated identification comparison but leaves open how identification behaves alongside rewriting.

Rewriting introduces the additional requirement of preserving the source meaning while making only the changes appropriate to the requested variety. Early BP-to-EP work treated adaptation as a dedicated conversion problem [Marujo et al. 2011], while linguistic analyses show that localized EP/BP differences can extend beyond individual words to multiword, phrase-level, and morphosyntactic alternatives [Barreiro and Mota 2018]. This range of possible changes supports a sequence-to-sequence formulation without implying that every input requires structural modification. Costa-jussà et al. [2018] show that a neural sequence-to-sequence system improves over a phrase-based approach in both Portuguese-variety directions and that human evaluators favor the neural outputs. More recently, Sanches et al. [2024] provide the closest dedicated rewriting comparison by evaluating pretrained BP-to-EP systems on the manually curated Golden Collection. Their experiments show that modern models can perform competitively on the task. Their benchmark's conservative references make unchanged-source copying an important reference for judging required edits. Portuguese identification and rewriting therefore have strong separate foundations, while less is known about how the two tasks behave when learned in the same model under different forms of variety-specific supervision.

2.2 Language Model Adaptation for Controlled Generation

Outside Portuguese, Khered et al. [2025] provide a methodological precedent by combining Arabic dialect identification and dialect-to-Modern-Standard-Arabic translation within a multi-task encoder-decoder model. Their experiments show that the shared setup, together with the additional training data used in the study, improves translation over the evaluated baselines. The reported gain supports the combined setup rather than multi-task learning.

Sharing the tasks addresses the model interface, while a separate question is how supervision of different scale and quality should be introduced during adaptation. Work on staged Machine Translation (MT) distinguishes broad exposure from smaller amounts of high-quality parallel data or targeted supervised specialization [Alves et al. 2024; Xu et al. 2024a]. The Tower+ and Omnilingual MT models extend this progression with preference, reward-based, or other post-training phases [Omnilingual MT Team et al. 2026; Rei et al. 2026]. Across these systems, broad adaptation, targeted supervision, and post-training serve distinct functions, supporting their separation when the available data differ in scale and quality.

Reward-based updates have also been used to refine already supervised translation models in scalar-feedback and low-resource dialect settings [Nguyen et al. 2017; Thu et al. 2023], establishing that task-specific feedback can guide post-training continuation.

Task-specific feedback can be defined through reference-free quality estimates, learned translation preferences, or fine-grained error information, each of which can provide task-relevant MT rewards [He et al. 2024; Ramos et al. 2026; Xu et al. 2024b]. However, reference-free reward optimization can become misaligned with actual translation quality [He et al. 2024], making reward design central. Recent systems therefore combine or decompose rewards across complementary criteria [Feng et al. 2025; Wang et al. 2026]. For EP/BP rewriting, the reward signal must favor required variety-specific changes without encouraging broader reformulation, while preserving the source meaning.

3 Experimental Setup

This section aims to define the experimental setup used to study Portuguese variety identification and rewriting. Section 3.1 introduces the data resources and explains how aligned EP/BP pairs are converted into task-specific rewriting and identification examples, Section 3.2 describes the model scales and control formulations, and Section 3.3 presents the staged training procedure and optimization settings. Section 3.4 defines supervised protocol selection, and Section 3.5 describes evaluation.

3.1 Data and Task Construction

The experiments draw on four aligned EP/BP resources whose different characteristics support broad adaptation, targeted specialization, and evaluation. OpenSubtitles [Tiedemann 2009] supplies large-scale exposure through aligned movie and television dialogue. Its scale supports broad EP/BP adaptation, although its pairs are noisier and less controlled than the targeted resources.

More controlled supervision comes from GPT-Wikipedia, a resource constructed in this work using Portuguese Wikipedia passages as topical inspiration for generated EP/BP pairs. The generation process is designed to preserve equivalent meaning while keeping the two varieties as close as possible, favoring localized lexical or morphosyntactic changes and minimizing edits not required for variety adaptation. Human review is used to distinguish appropriate EP/BP contrasts from unnecessary changes, while equal pairs are retained as copying supervision when no change is needed.

FRMT adds region-aware variation through a machine translation resource and benchmark derived from English source sentences with regional outputs, including EP and BP [Riley et al. 2023]. FRMT is used primarily as a held-out evaluation resource, while its training split is added only in the GPT-Wikipedia + FRMT condition to examine how this additional supervision changes identification and rewriting performance. The Golden Collection complements FRMT with a manually curated rewriting benchmark used only for evaluation [Sanches et al. 2024]. Many references closely follow the source and favor conservative rewriting.

Their relative scales and split roles are summarized before task construction in Table 1.

Table 1. EP/BP pair counts by resource and split before task construction.

Resource	Train	Validation	Test
OpenSubtitles	10,347,883	0	0
GPT-Wikipedia	4,519	564	566
FRMT	2,502	20	2,611
Golden Collection	0	0	500

The OpenSubtitles counts describe the processed base resource, which has no separate validation split. A 200-instance validation view is sampled from its processed training data when the Stage A data are prepared, whereas the FRMT training split supplies the additional Stage B supervision and its held-out test split is used for evaluation. These resource-level counts precede task construction. Each aligned pair yields one BP-to-EP and one EP-to-BP rewriting example, while a non-equal pair also yields one identification example for each variety. Equal pairs remain useful for rewriting but are excluded from identification, since identical text cannot unambiguously support two opposing labels.

3.2 Models and Control Formulations

The experiments use T5Gemma 2 encoder-decoder models with 270M and 4B parameters [Zhang et al. 2025]. Their text-to-text architecture provides a single interface for short variety-label outputs and full-sentence rewriting. The 270M-parameter model is fully fine-tuned in a lower-cost supervised protocol-selection study that defines the training protocol used in the subsequent experiments. Those experiments use LoRA [Hu et al. 2022] with rank 48, $\alpha = 96$, and dropout 0.05, yielding 187,772,928 trainable parameters while freezing the 4B-parameter base model.

Across both tasks, `PT` labels an EP source sentence and `BR` labels a BP source sentence. The two control formulations differ in where this input-variety information is represented. In encoder-controlled rewriting, the relevant label prefixes the encoder input and the decoder target contains only the rewritten sentence. Encoder-controlled identification instead introduces `CLS` before the input sentence, while the decoder predicts `PT` or `BR`.

The decoder-controlled formulation moves the input-variety label to the generated sequence. Its rewriting inputs contain only the source sentence, while each target begins with `PT` or `BR` and continues with the rewrite. For identification, the same unprefixed input is paired with a label-only target. For decoder-controlled identification, loss is computed only on the `PT` or `BR` label token, whereas rewriting uses the complete target sequence. Section 3.4 specifies task mixing, tokenizer treatment, and label scoring.

3.3 Staged Training

Supervised adaptation uses the identification and rewriting examples described above, progressing from large-scale OpenSubtitles exposure in Stage A to targeted specialization in Stage B. In Stage A, OpenSubtitles exposes the models to large-scale aligned EP/BP supervision. After task construction, the Stage A pool contains 20,695,766 rewriting examples and 7,666,883 identification examples. The fixed optimization budgets expose the models to only a small fraction of this pool, so Stage A examples do not repeat before the maximum step budget is reached. Stage A's contribution is isolated with matched configurations that start directly at Stage B under the same specialization setup.

Stage B moves from large-scale subtitle exposure to targeted supervised specialization. GPT-Wikipedia provides the primary targeted condition, while the GPT-Wikipedia + FRMT condition adds the complete FRMT training split to examine how additional region-aware supervision changes identification and rewriting. The two training splits are concatenated without weighting or resampling, while the task templates and training procedure remain unchanged. After task construction, the GPT-Wikipedia condition contains 9,038 rewriting examples and 6,652 identification examples. The GPT-Wikipedia + FRMT condition contains 14,042 rewriting examples and 11,616 identification examples. For the 4B-parameter model, both Stage B training sets are smaller than the maximum training-example presentation budget and can therefore repeat across shuffled epochs. For the 270M model, this budget exceeds GPT-Wikipedia but not GPT-Wikipedia + FRMT.

Stage C retains both Stage B data conditions, GPT-Wikipedia and GPT-Wikipedia + FRMT, but continues each Stage B checkpoint using a more conservative rewriting-only subset. The subsets are selected to favor localized EP/BP changes, structurally close pairs, and unchanged examples where copying is appropriate. The GPT-Wikipedia Stage C subset contains 6,686 rewriting examples balanced across directions. The GPT-Wikipedia + FRMT Stage C subset contains 6,828 balanced rewriting examples, comprising the same 6,686 GPT-Wikipedia examples and 142 retained FRMT examples. This combined set underlies the Stage C rows for the GPT-Wikipedia + FRMT condition in the result tables.

For each Stage C input, four candidate rewrites are sampled and the reference rewrite is added to the group. Candidate rewards are compared within this group to form the policy update. For each candidate, the BLEU [Papineni et al. 2002] and TER [Snover et al. 2006] components compare its agreement with the reference against the score obtained by returning the source unchanged, making the reward relative to a copy baseline rather than to the reference score alone. Under encoder control, the reward assigns equal weight to the relative BLEU and TER components, $0.5R_{\text{BLEU}} + 0.5R_{\text{TER}}$. Under decoder control, it uses $0.4R_{\text{BLEU}} + 0.4R_{\text{TER}} + 0.2R_{\text{label}}$, where R_{label} rewards a correct first `PT`/`BR` label. After removing any decoder control label and normalizing whitespace, 0.10 is subtracted when a candidate copies the source even though the reference requires a change.

Table 3 summarizes these optimization settings.

All five settings use AdamW [Loshchilov and Hutter 2019] with a linear learning-rate schedule, with weight decay set to 0.01 in supervised Stages A and B and to 0 in Stage C. Supervised Stages A and B stop early and restore the best checkpoint.

3.4 Supervised Protocol Selection

Four supervised protocols are evaluated with the fully fine-tuned 270M-parameter decoder-controlled model. All four variants follow the same OpenSubtitles Stage A and GPT-Wikipedia Stage B path, with the same T5Gemma 2 checkpoint, task construction, and comparable step budgets, and are evaluated with the common procedure described in Section 3.5. Only the supervised protocol component changes, and the selected protocol is then used for the 4B-parameter experiments.

The base protocol combines rewriting and identification rows in the same shuffled training mixture, so their contribution follows their natural frequency rather than an enforced task balance. `PT`

Table 2. Sequence-to-sequence task templates. E and D denote encoder-controlled and decoder-controlled formulations, respectively. x_{PT} and x_{BR} are aligned EP and BP sentences, while x is an input sentence whose variety is to be identified.

Setup	Direction	Encoder input	Decoder target
E rewriting	BP-to-EP	BR x_{BR}	x_{PT}
	EP-to-BP	PT x_{PT}	x_{BR}
E identification	—	CLS x	PT or BR
D rewriting	BP-to-EP	x_{BR}	BR x_{PT}
	EP-to-BP	x_{PT}	PT x_{BR}
D identification	—	x	PT or BR

Table 3. Optimization settings by stage and model (FT = full fine-tuning).

Stage	Model / adaptation	Learning rate	Warm-up steps	Maximum steps
A	270M full FT	2×10^{-5}	200	2,500
A	4B LoRA	1×10^{-5}	400	2,500
B	270M full FT	2×10^{-5}	150	1,200
B	4B LoRA	1×10^{-5}	150	1,200
C	4B LoRA	5×10^{-6}	20	600

and BR use their existing base-vocabulary representations rather than newly registered tokenizer tokens. For identification examples, loss is computed at the supervised label position against the normal decoder vocabulary, while rewriting uses the complete target sequence and normal decoder vocabulary. The task-balanced variant changes only batch composition by enforcing equal rewriting and identification representation in each optimizer update. The added-token variant keeps the same control strings but explicitly registers PT and BR with the tokenizer. The restricted-classification-loss variant changes only identification scoring by restricting the competing output labels to PT and BR, while rewriting remains unchanged. The results of this comparison are presented in Section 4.1.

3.5 Evaluation

The two tasks are evaluated under both control formulations on FRMT and the Golden Collection. Accuracy and macro-F1 measure identification, while BLEU [Papineni et al. 2002], computed with SacreBLEU [Post 2018], and TER [Snover et al. 2006] measure rewriting. The principal aggregate rewriting comparisons use corpus BLEU and TER, reported separately for BP-to-EP and EP-to-BP, whereas sentence-level BLEU and TER are reserved for diagnostic analyses and also contribute to the Stage C reward described in Section 3.3. Rewriting uses beam search with beam size 4. Before the rewriting scores are computed, control labels are removed from predictions and references so that the metrics evaluate sentence content rather than the control tokens. The copy baseline returns the source unchanged and provides an informative rewriting reference when no modification is required. Section 4.4 examines this behavior with representative outputs.

4 Results

This section aims to determine how supervised protocol choice, control placement, training-data composition, staged adaptation, and reward-based continuation affect identification and rewriting. Section 4.1 reports the protocol-selection results, Section 4.2 compares supervised identification and rewriting after Stage B, and Section 4.3 examines reward-based continuation. Section 4.4 analyzes representative outputs and copying behavior.

4.1 Supervised Protocol Selection

The 270M-parameter protocol-selection study evaluates which supervised protocol provides the most suitable basis for the subsequent experiments with the 4B-parameter model. Table 4 compares the four candidates after the same OpenSubtitles Stage A and GPT-Wikipedia Stage B path.

No candidate consistently improves identification and rewriting across both evaluation resources. The alternatives instead improve isolated metrics. Task-balanced batches improve Golden Collection identification while reducing the other reported scores. Added-token treatment improves FRMT identification but not rewriting or Golden Collection identification. Restricted loss gives the strongest Golden Collection values and a small FRMT BLEU increase, but it does not improve FRMT identification. Since these improvements are confined to individual metrics while requiring additional batching, tokenizer, or loss mechanisms, the base protocol is retained as the simpler, broadly competitive configuration for the subsequent experiments with the 4B-parameter model. The retained protocol therefore uses the natural rewriting-to-identification mixture, the existing PT/BR base-vocabulary representations, and normal decoder-vocabulary scoring at the supervised identification label position, while rewriting uses loss over the complete target sequence.

4.2 Supervised Identification and Rewriting Results

With the supervised protocol selected by the 270M-parameter study, the 4B-parameter T5Gemma 2 model is adapted with rank-48 LoRA under both control formulations. Tables 5 and 6 report identification and rewriting results after Stage B and Stage C.

Stage B identification shows how adding FRMT supervision changes the two evaluation resources. On FRMT, macro-F1 rises from 82.39 to 86.06 under encoder control and from 81.68 to 86.10 under decoder control. On the Golden Collection, the effect is mixed: encoder-controlled macro-F1 decreases from 73.92 to 71.19, while decoder-controlled macro-F1 increases from 71.53 to 72.38. Adding FRMT improves FRMT identification under both formulations but gives no consistent Golden Collection gain, and neither control formulation is consistently stronger overall.

For external context, the encoder-controlled GPT-Wikipedia Stage B configuration reaches 82.39 macro-F1 on FRMT, compared with the 77.25 F1 reported by Sousa et al. [2025] for their strongest BERTimbau-based identifier. Although both studies use FRMT for evaluation, the comparison is contextual, as Sousa et al. [2025] use

Table 4. Decoder-controlled 270M full-fine-tuning protocol-selection results. Macro-F1 is a percentage, BLEU uses a 0–100 scale, and lower TER is better.

Protocol variant	FRMT					Golden Collection				
	Macro-F1 (%)	BLEU (0–100)		TER		Macro-F1 (%)	BLEU (0–100)		TER	
		BP-to-EP	EP-to-BP	BP-to-EP	EP-to-BP		BP-to-EP	EP-to-BP	BP-to-EP	EP-to-BP
Base protocol	70.76	38.56	38.69	**0.4748**	**0.4828**	67.63	86.45	86.70	0.0978	0.0981
Task-balanced batches	69.74	38.56	38.55	0.4751	0.4859	68.52	86.04	86.45	0.0987	0.1001
Added-token path	**72.93**	38.36	38.67	0.4768	0.4864	66.66	86.23	86.60	0.0986	0.1006
Restricted classification loss	70.74	**38.70**	**38.72**	0.4750	0.4832	**69.10**	**87.01**	**87.32**	**0.0941**	**0.0956**

Table 5. Identification results for the 4B-parameter T5Gemma 2 model with rank-48 LoRA.

Stage	Control	Data condition	FRMT		Golden Collection	
			Accuracy (%)	Macro-F1 (%)	Accuracy (%)	Macro-F1 (%)
B	Encoder	GPT-Wikipedia	82.39	82.39	**73.99**	**73.92**
B	Decoder	GPT-Wikipedia	81.69	81.68	71.61	71.53
B	Encoder	GPT-Wikipedia + FRMT	86.09	86.06	71.43	71.19
B	Decoder	GPT-Wikipedia + FRMT	**86.15**	**86.10**	72.53	72.38
C	Encoder	GPT-Wikipedia	82.33	82.33	73.63	73.50
C	Decoder	GPT-Wikipedia	81.52	81.31	72.34	71.81
C	Encoder	GPT-Wikipedia + FRMT	85.82	85.78	71.43	71.17
C	Decoder	GPT-Wikipedia + FRMT	85.65	85.54	72.34	72.02

a dedicated identification model with a different training setup and objective, while the present configuration learns identification and rewriting in the same model.

Across the Stage B comparisons, data composition produces larger changes in rewriting than in identification. Adding FRMT changes FRMT rewriting differently across directions and control formulations, ranging from almost no change in some comparisons to a clear improvement in others. In contrast, Golden Collection BLEU drops by roughly 20 points across every direction and control comparison, with corresponding increases in TER.

The rewriting-only baselines test whether identification supervision harms rewriting. With GPT-Wikipedia, the multitask and rewriting-only configurations remain very close: multitask BLEU is higher in three of the four dataset-direction comparisons, while rewriting-only is higher only on FRMT EP-to-BP, by 0.21 BLEU. With GPT-Wikipedia + FRMT, the multitask configuration obtains higher BLEU in all four comparisons, including gains of 9.04 points in Golden Collection BP-to-EP and 7.92 points in EP-to-BP. Across the eight directional BLEU comparisons, rewriting-only is higher only once and by a small margin, so removing identification supervision does not improve rewriting overall.

Control placement shows a clearer directional tendency: decoder control leads every BP-to-EP comparison, while EP-to-BP is mixed and sometimes favors encoder control.

The most directly comparable prior rewriting result is BP-to-EP evaluation on the Golden Collection. After rescaling the BLEU values reported by Sanches et al. [2024] to the 0–100 scale, their strongest trained system obtains approximately 84 BLEU and 0.10 TER, while their published single-direction copy baseline reaches approximately 89 BLEU and 0.07 TER. The decoder-controlled GPT-Wikipedia configuration reaches 87.75 BLEU and 0.0874 TER after Stage B and 88.44 BLEU and 0.0824 TER after Stage C. The comparison is contextual, as Sanches et al. [2024] train specifically for BP-to-EP rewriting, while the present configuration is bidirectional and also learns identification. For broader external context, AMALIA SFT and DPO [Simplício et al. 2026], Qwen3-4B [Yang et al. 2025], and FLAN-T5-XL [Chung et al. 2024] are evaluated descriptively without task-specific fine-tuning on the EP/BP resources used in this work. Across these external systems, the strongest adapted T5Gemma 2 result is stronger in every reported setting except FRMT BP-to-EP, where AMALIA SFT obtains 44.65 BLEU and 0.4289 TER versus 43.97 BLEU and 0.4338 TER.

Finally, Table 7 isolates the contribution of Stage A OpenSubtitles warm-up by comparing each configuration with its corresponding no-Stage-A variant. For configurations whose Stage B uses GPT-Wikipedia, Stage A shows its clearest benefit under encoder control, especially on the Golden Collection. On FRMT, Stage A adds 0.87 macro-F1 points and 4.40 BLEU points while reducing TER by 0.0573. On the Golden Collection, the same configuration gains 4.01 macro-F1 points and 25.41 BLEU points while reducing TER by 0.2404. Under decoder control with GPT-Wikipedia, Stage A has almost no rewriting effect on FRMT and lowers macro-F1 by 1.28 points, but improves all three aggregate measures on the Golden Collection. The GPT-Wikipedia + FRMT conditions show smaller or mixed changes: encoder control improves FRMT but lowers Golden Collection rewriting by 6.79 BLEU points, whereas decoder control produces small rewriting gains on both resources.

4.3 Reward-Based Continuation

Tables 5 and 6 also report how the Stage B configurations change after Stage C. Although continuation uses rewriting-only data, identification remains close to the corresponding Stage B

Table 6. Directional rewriting results for the 4B-parameter T5Gemma 2 model after Stage B and Stage C with rank-48 LoRA. Lower TER is better.

Stage	Control	Data condition	FRMT		Golden Collection	
			BP-to-EP BLEU (0–100) / TER	EP-to-BP BLEU (0–100) / TER	BP-to-EP BLEU (0–100) / TER	EP-to-BP BLEU (0–100) / TER
B	Encoder	GPT-Wikipedia	37.87/0.4791	40.10/0.4714	86.08/0.0946	**89.09**/**0.0823**
B	Encoder (rewriting-only)	GPT-Wikipedia	37.73/0.4803	40.31/0.4715	85.66/0.0976	88.66/0.0847
B	Decoder	GPT-Wikipedia	40.45/0.4594	40.60/0.4673	87.75/0.0874	87.50/0.0916
B	Encoder	GPT-Wikipedia + FRMT	37.99/0.4770	**45.43**/0.4273	65.14/0.2555	66.61/0.2535
B	Encoder (rewriting-only)	GPT-Wikipedia + FRMT	36.76/0.4852	45.40/**0.4251**	56.10/0.3222	58.69/0.3103
B	Decoder	GPT-Wikipedia + FRMT	**43.97**/**0.4338**	44.74/0.4306	68.37/0.2336	67.83/0.2400
C	Encoder	GPT-Wikipedia	37.89/0.4784	40.28/0.4696	86.22/0.0953	88.93/0.0837
C	Decoder	GPT-Wikipedia	40.34/0.4603	40.33/0.4698	**88.44**/**0.0824**	88.02/0.0878
C	Encoder	GPT-Wikipedia + FRMT	38.35/0.4756	43.47/0.4440	77.16/0.1601	79.95/0.1488
C	Decoder	GPT-Wikipedia + FRMT	42.72/0.4417	42.72/0.4466	81.52/0.1351	80.70/0.1402

Table 7. Stage A warm-up ablation under aggregate evaluation. Deltas are computed as the Stage A model minus the corresponding no-Stage-A model. Positive deltas indicate improvement for macro-F1 and BLEU, while negative deltas indicate improvement for TER.

Control	Data condition	Dataset	Δ Macro-F1 (pp)	Δ BLEU (points)	Δ TER
Encoder	GPT-Wikipedia	FRMT	**+0.87**	**+4.40**	**-0.0573**
Encoder	GPT-Wikipedia	Golden Collection	**+4.01**	**+25.41**	**-0.2404**
Encoder	GPT-Wikipedia + FRMT	FRMT	+0.81	+1.92	-0.0176
Encoder	GPT-Wikipedia + FRMT	Golden Collection	+1.62	-6.79	+0.0425
Decoder	GPT-Wikipedia	FRMT	-1.28	-0.01	+0.0004
Decoder	GPT-Wikipedia	Golden Collection	+2.67	+1.42	-0.0144
Decoder	GPT-Wikipedia + FRMT	FRMT	-0.07	+0.35	-0.0031
Decoder	GPT-Wikipedia + FRMT	Golden Collection	+2.19	+2.00	-0.0217

values. Macro-F1 changes are small across resources and controls, with increases and decreases rather than a consistent gain.

The rewriting effect depends strongly on the Stage B supervision condition. After GPT-Wikipedia specialization, Stage C changes rewriting only modestly, with BLEU differences below one point across all four dataset-direction comparisons. The larger effect follows GPT-Wikipedia + FRMT specialization. Golden Collection BLEU recovers by approximately 12 points in both directions under both control formulations, while FRMT rewriting changes are smaller and differ across directions and formulations.

Since continuation changes both the optimization procedure and the rewriting subset used after Stage B, the reported comparison does not separate the effect of reward-based optimization from the effect of the continuation data.

4.4 Qualitative Analysis

Source-reference similarity and model copying behavior help explain the rewriting differences observed above. The Golden Collection contains a high proportion of exact or near-copy references. Of its 1,000 bidirectional rows, 454 are exact source-reference matches, 566 have copy TER at most 0.05, and 636 have copy TER at most 0.08. Consistent with this composition, the unchanged-source baseline reaches 89.17 BLEU and 0.0755 TER on the Golden Collection. The corresponding FRMT baseline is substantially weaker, at 38.75 BLEU and 0.4819 TER, consistent with larger source-reference differences in that resource. This contrast helps explain why unnecessary rewriting can be costly on the Golden Collection while FRMT provides more room to improve over unchanged-source copying.

These reference profiles also clarify why useful local adaptations do not necessarily raise corpus BLEU above the Golden Collection copy baseline. A required edit can improve only a limited span, whereas an unnecessary paraphrase can remove many matching higher-order n-grams. Aggregate BLEU can therefore favor the unchanged source even when local edits are appropriate, but this does not make copying universally desirable.

Encoder-controlled Stage B models make this contrast between conservative and more extensive rewriting visible. On the 454 exact-copy rows, the model trained with GPT-Wikipedia data preserves the source in 74.7% of cases, compared with 18.7% for the model trained with the GPT-Wikipedia + FRMT data mix. On these rows, preserving the source is consistent with the reference. The model trained with GPT-Wikipedia data also remains more conservative on the 546 non-exact rows, copying the source in 33.7% of cases compared with 3.1% for the model trained with the GPT-Wikipedia + FRMT data mix. On these rows, copying can indicate under-editing when the reference contains a required variety-specific change, although a source-reference difference can also reflect paraphrastic or editorial changes that are not required for EP/BP adaptation. Supervision from the GPT-Wikipedia + FRMT data mix sharply reduces copying, which can help realize

required changes but also creates more opportunities for unnecessary reformulation. Table 8 illustrates how the two Stage B supervision conditions translate into different rewriting behavior.

In the first example, the reference replaces *procura* with *demanda*. The model trained with GPT-Wikipedia data leaves the source unchanged despite the required lexical edit, whereas the model trained with the GPT-Wikipedia + FRMT data mix makes the change and reformulates the sentence more broadly. In the second example, the reference is identical to the source. The model trained with GPT-Wikipedia data preserves it, while the model trained with the GPT-Wikipedia + FRMT data mix unnecessarily reformulates the unchanged sentence.

A separate issue concerns the references used for evaluation. Some source-reference pairs contain differences that extend beyond localized EP/BP conversion, so a model can produce an appropriate variety-specific rewrite without reproducing every transformation present in the reference. Table 9 gives representative cases from the Golden Collection and FRMT. In the Golden Collection example, the reference removes the final clause describing the returned gesture, while in the FRMT example it removes the feet conversions while preserving the corresponding measurements in metres. Other inspected FRMT references contain lexical substitutions such as *marcante* to *importante*, *ganhou* to *recebeu*, *reside* to *vive*, *usar* to *utilizar*, and *música* to *canção*, which are valid alternatives, not consistently required for variety adaptation.

5 Discussion and Limitations

Taken together, the quantitative and qualitative results show that aligned EP/BP resources are not interchangeable simply by virtue of containing parallel sentences. Supervision composition produces substantially larger changes in rewriting than in identification, and Table 8 shows that the GPT-Wikipedia and GPT-Wikipedia + FRMT conditions induce markedly different rewriting behavior. GPT-Wikipedia was constructed around localized variety adaptation, while FRMT provides a broader region-aware translation resource. Under the evaluated conditions, the usefulness of aligned supervision depends on how closely training-pair transformations match the intended rewriting behavior, not merely on parallel-data volume.

Control placement has a smaller and less uniform effect than supervision composition. Decoder control is consistently stronger in the reported BP-to-EP rewriting comparisons, whereas EP-to-BP remains mixed and includes settings where encoder control performs better. Identification likewise provides no universal ordering between the two formulations across the evaluation resources. The experiments establish these differences but do not determine why encoder- or decoder-side variety information changes relative performance, so control placement remains an empirical choice with no uniformly preferable formulation.

For configurations whose Stage B uses GPT-Wikipedia, Stage A provides its clearest aggregate benefit on the Golden Collection. The Golden Collection's conservative references are closely aligned with the localized rewriting behavior targeted by GPT-Wikipedia, while the other evaluated conditions show smaller or mixed changes. Stage C changes rewriting only modestly after GPT-Wikipedia specialization but substantially restores Golden Collection performance after GPT-Wikipedia + FRMT specialization, with smaller and mixed changes on FRMT. These patterns indicate that an additional training stage modifies a rewriting policy established by the preceding supervision rather than providing a uniform gain across evaluation resources.

Rewriting evaluation is limited by both the construction of the reference pairs and the automatic metrics used to compare outputs with them. As the source-reference examples in Section 4.4 show, some evaluation pairs contain paraphrastic, editorial, or information-level differences beyond the localized variety changes required by the task, while the Golden Collection also contains many cases that require little or no rewriting. Even with an appropriate reference, BLEU and TER remain surface-oriented measures, and a single reference cannot reliably distinguish every required variety change, appropriate copying, acceptable alternative phrasing, and unnecessary reformulation.

Stage C inherits the same reference dependence. For each candidate, sentence-level BLEU and TER are computed against the reference and compared with the scores obtained by returning the source unchanged. These relative scores then contribute to the reward.

All reported experiments use the T5Gemma 2 encoder-decoder family, so the effects of supervision composition, control placement, and staged adaptation have not been established across multiple model families. The 270M-parameter full-fine-tuning study and the experiments with the 4B-parameter model serve different purposes and use different adaptation methods, so they do not constitute a controlled comparison of scale or fine-tuning strategy. Stage C also changes the continuation subset and optimization procedure together, which prevents their individual contributions from being separated in the reported comparison.

6 Conclusion

This work examined Portuguese variety identification and bidirectional EP/BP rewriting within a shared T5Gemma 2 text-to-text setup, with particular attention to how supervision composition, control placement, and staged adaptation shape the two tasks. The experiments show that one adapted model can support both identification and rewriting, while rewriting behavior is especially sensitive to the transformations represented in the aligned supervision. GPT-Wikipedia, which was constructed around localized variety adaptation, produces conservative rewriting that aligns closely with the Golden Collection, whereas adding FRMT changes the learned rewriting behavior substantially and produces different effects across resources and directions.

Control placement and staged adaptation refine this picture without providing a single universally preferable configuration. Adding FRMT consistently improves FRMT identification under both control formulations, while its effect on Golden Collection identification is mixed. For rewriting, decoder control is consistently stronger for BP-to-EP, whereas EP-to-BP remains mixed across the evaluated conditions, and removing identification supervision does not improve rewriting overall. Stage A provides its clearest benefit in selected GPT-Wikipedia conditions, while Stage C changes the GPT-Wikipedia condition only modestly and has its largest effect after GPT-Wikipedia + FRMT specialization, where it restores a substantial part of the Golden Collection rewriting performance. The adapted configurations also numerically exceed the closest published task-specific results used for comparison, namely Sousa et al. [2025] on FRMT identification and the strongest trained Sanches et al. [2024] system on Golden Collection BP-to-EP rewriting, although these comparisons involve dedicated setups that differ from the shared bidirectional setting

Table 8. Representative Golden Collection outputs from encoder-controlled Stage B models trained with GPT-Wikipedia and GPT-Wikipedia + FRMT. Boldface marks model-output spans illustrating under-editing or unnecessary reformulation.

Source	Reference	GPT-Wikipedia	GPT-Wikipedia + FRMT
"A procura aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP).	*"A demanda aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP).*	*"A **procura** aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP).*	*"A demanda **está aumentando drasticamente em todo o mundo**", **disse** o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia em São Carlos (SP).*
Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.	*Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.*	*Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.*	***Por isso, o parecer do ilustre Relator, Bernardo Cabral, é absolutamente correto tanto no conteúdo como na forma.***

Table 9. Examples where source and reference sentences differ in expressed information beyond localized variety conversion. Boldface marks source spans whose information is omitted from the reference.

Dataset	Direction	Source	Reference
Golden Collection	BP-to-EP	*- Oww você é um amor Obrigado viu!- **Falei e o mesmo retribuiu com um abraço.***	*- Oww, és adorável. Obrigado!*
FRMT	BP-to-EP	*O Autódromo Internacional Orlando Moura tem uma pista de 3.433 metros **(11.263 pés)**, e o Kartódromo Ayrton Senna uma pista de 930 metros **(3.051 pés)**.*	*O Autódromo Internacional Orlando Moura tem uma pista de 3433 metros e o Kartódromo Ayrton Senna tem uma pista de 930 metros.*

used in this work. Across the additional external systems considered, the strongest adapted T5Gemma 2 result reported in this work is stronger in every reported identification and rewriting setting except FRMT BP-to-EP, which favors AMALIA SFT.

Effective EP/BP rewriting depends both on supervision whose transformations reflect the intended variety-adaptation task and on evaluation resources and metrics that can distinguish required edits from appropriate preservation and acceptable alternative phrasing. Future work should therefore prioritize more tightly verified variety-focused supervision together with human evaluation, targeted linguistic checks, multiple acceptable references, or validated learned evaluators that better capture these distinctions. Further experiments should separate continuation-data selection from reward-based optimization, extend the comparison to other model families and domains, and include a frontier proprietary model under the same identification and bidirectional rewriting protocol.

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